

National University of Computer and Emerging Sciences, Lahore Campus



Course:
Program:
Submission Date:
Section:
Assignment no. :
Total Marks:

Data Mining
BSE, BCS
21-04-26
6A, 8A
03
30

Important Instructions:

- Use A4 sheets or narrow line sheets for solving assignments.
- Assignments done on register pages will be cancelled.
- Must attach the assignment title page shared to you at the end of assignment.
- This assignment must be completed independently by the student. The use of AI tools (including ChatGPT, Bard, Copilot, or any other generative AI system) is strictly prohibited.
- Students are expected to show all manual calculations and intermediate steps clearly.

Question 1: Ripper Algorithm

(20 Marks)

ID	Age	Income	Student	Credit Rating	Online Shopper	Buys Laptop
1	<=25	High	No	Fair	No	No
2	<=25	High	No	Excellent	No	No
3	26-35	High	No	Fair	Yes	Yes
4	>35	Medium	No	Fair	Yes	Yes
5	>35	Low	Yes	Fair	Yes	Yes
6	>35	Low	Yes	Excellent	No	No
7	26-35	Low	Yes	Excellent	Yes	Yes
8	<=25	Medium	No	Fair	No	No
9	<=25	Low	Yes	Fair	Yes	Yes
10	>35	Medium	Yes	Fair	Yes	Yes
11	<=25	Medium	Yes	Excellent	Yes	Yes
12	26-35	Medium	No	Excellent	Yes	Yes
13	26-35	High	Yes	Fair	Yes	Yes
14	>35	High	No	Excellent	No	No

Tasks:

Apply RIPPER Algorithm:

1. Growing Phase (calculate Information Gain)
2. Pruning Phase (calculate Accuracy)
3. Optimization Phase
4. Final Rule Set

Formulas

Information Gain: $\text{Gain} = p * (\log(p/(p+n)) - \log(P/(P+N)))$

Accuracy: $\text{Accuracy} = (p - n) / (p + n)$

Test Data

1. Age ≤ 25 , Income = Medium, Student = Yes, Credit = Fair, Online Shopper = Yes
2. Age > 35 , Income = High, Student = No, Credit = Excellent, Online Shopper = No

Question 2: Comparison of Classification Approaches (10 Marks)

Compare Rule-Based Classification (**RIPPER**) with Decision Tree (e.g., **C4.5** Decision Tree Algorithm).

Must discuss following points briefly:

1. Model Representation
2. Learning Strategy
3. Handling Noise
4. Interpretability (Which is easier to understand and why?)
5. Overfitting Control (Compare pruning mechanisms)
6. Real-world suitability (When would you prefer RIPPER over Decision Trees?)

National University of Computer and Emerging Sciences

Department of Computer Science, School of Computing

DATA MINING



Assignment # _____

Section: _____

Submitted by:

Name: _____

Roll no: _____

Submitted to: Dr. Muhammad Aasim Qureshi

Date Assigned: _____

Date Submission: _____